(c) Western disturbances (d) South-West Monsoon

U.P. P.C.S. (Pre) 2021

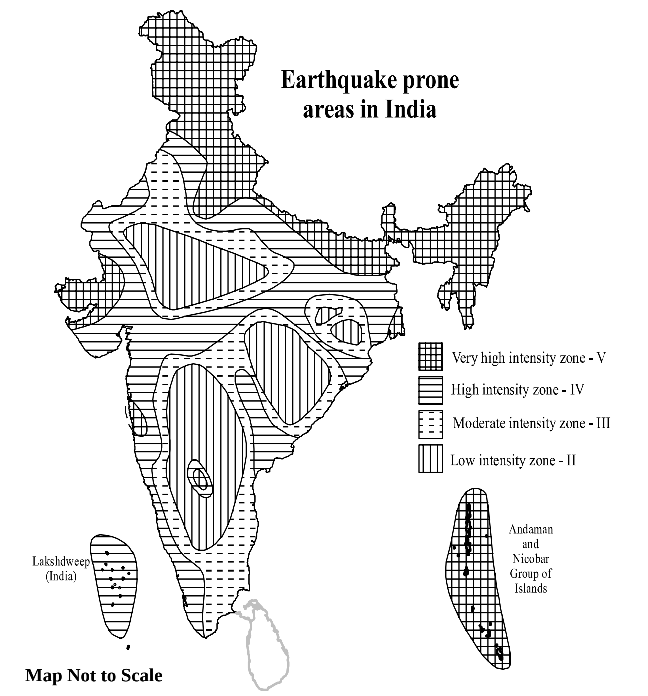
Ans. (c)

The western disturbance is responsible for the winter rains in the northwest part of India, but the western disturbance is a weak form of cyclonic disturbance. In this form, it also comes under cyclonic depression, but the western disturbance will be the correct answer as the most suitable option.

Natural Disasters

Natural disasters are catastrophic events of atmospheric, geological and hydrological origin that can cause fatalities, property damage and socio-environmental disruption. *Volcanic eruptions, Earthquakes, Oceanic Vibrations, Droughts, Floods, Cyclones, Soil erosion, Snow avalanches, Mud Flows, etc. are example of Natural Disasters. Among these **most of the disasters are produced by Natural forces.** However, unwanted activities of human's increase the intensity of these disasters. Due to the exploitative activities of human beings, the adverse effects of these disasters have increased.

*The Himalayan mountain is a new fold mountain whose growth is still continuing. The collision of the Indian Plate along the boundary of Nepal formed the orogenic belt that created the Himalayan Mountains. **The Indian Plate is moving faster than the Eurasian Plate** in a northward direction. Hence, the associated compression due to this phenomenon has kept the Himalayas growing in size.



*The Southern Plateau is relatively more stable, yet the rare events of low intensity earthquakes in the region raise questions about the stability of Peninsular India.

*On the basis of data provided by the India Meteorological Department and other agencies, the Bureau of Indian Standards has published the latest version of seismic zone map of India. The Bureau of Indian standards has prepared **'Earthquake Resistant Design Code of India, IS - 1893, Part I : 2002** which is a new edition of seismic zone mapping of India. In this, India has been divided into four seismic zones on the basis of earthquake intensity - namely **Zone II, Zone III, Zone IV** and **Zone V**. Each zone indicates the effects of an earthquake at a particular place based on the observations of the affected area and can also be described using a **descriptive scale like the Modified Mercalli Intensity** Scale or the **Medvedev-Sponhever-Karnik Scale (MSK).**

Seismic Zone	**Intensity**
Zone II (Low Intensity)	MSK VI or less, Low Damage Risk Zone
Zone III (Moderate Intensity)	MSK VII, Moderate Damage Risk Zone
Zone IV (High Intensity)	MSK VIII, High Damage Risk Zone
Zone V (Very High Intensity)	MSK IX or greater, Very High Damage Risk Zone

***Zone V:** Zone V covers **10.9% of the area of the country** with the highest risk zone that suffers earthquakes of intensity MSK IX or greater. The IS code assigns a zone factor of **0.36 for Zone V**. It is referred to as the very high damage risk zone. The region of J&K (UT), Ladakh (UT) some parts of Uttarakhand and Bihar, the North - East Indian region and the Rann of Kutch, Andaman & Nicobar etc., fall in this zone.

***Zone IV** - The IS code assigns a zone factor of **0.24 for Zone IV**. The Indo - Gangetic basin and the **National Capital Territory of Delhi**, as well as the rest of Himalayas, are included. in Zone IV. In Maharashtra, the Patan area (Koyananagar) falls under Zone IV.

This zone covers the northern parts of Uttar Pradesh, Bihar and West Bengal, parts of Gujarat and small portions of Maharashtra near the west coast and Rajasthan. It covers almost **17.3% of the areas of the country**.

*__Zone III__ - The IS code assigns a zone factor of **0.16 for Zone III**. It covers the the Lakshadweep Islands. This zone is classified as a moderate damage risk zone. **It covers 30.4% of the areas of the country.**

*__Zone II__- The IS code assigns zone factor of 0.10 for Zone II. Zone II is determined by combining the previous Zone I and Zone II. It **covers 41.4% of India.** Major parts of the peninsular region and Karnataka Plateau fall in this zone. Thus, it does not come under the zone of high seismic intensity.

*__Tsunami__ is a Japanese word, which consists of two words, **'tsu'** meaning 'harbours' and 'nami' meaning 'wave'. According to the High Powered Committee on Disaster Management Report (2001) of India, it originates from events like Earthquakes in the Ocean, Landslides, Volcanic explosions, etc. The **majority of Tsunamis originate from earthquakes** that occur on the ocean floor. Tsunami is also called the **'Harbour waves'** or '**Sea waves'**. The **Indian Tsunami Early Warning System (ITEWS)** is established at Indian National Centre for Ocean Information Science (INCOIS), **Hyderabad**. It started operating in October 2007. On 26th December, 2004, a Tsunami, that originated from the Indian Ocean, hit the Coromandel coast of India. The Coromandel coast is southeastern coast of the Indian subcontinent. The coastlines run from **False Divi Point in north** to Kanyakumari in the south.

*The **India Meteorological Department** was constituted in **1875** under the Ministry of Earth Sciences, with its headquarters at **New Delhi**. *The Indian Institute of Tropical Meteorology and the deputy director's office of IMD (India Meteorological Department) are situated in Pune. Kolkata was the first head office of India Meteorological Department. The headquarters of IMD was later shifted to **Shimla in 1905**, then to Pune in 1928, and finally to **Delhi in 1944**.

*Floods are an overflow of water submerging land is usually dry. The second important consequence of uncertainty and irregularity of monsoons is the flooding. ***"The Flood Forecast & Warning Organisation"** was set up by **Central Water Commission in 1969** to establish forecasting sites on Inter-State rivers in various flood prone areas of the country. *The **'National Flood Forecasting and Warning Network'** of Central Water Commission, comprises of 338 flood forecasting sites. *The flood region of Uttar Pradesh is divided into high flood prone and low flood prone regions.

*A **cyclone** is a large scale air mass that rotates around a strong centre of low pressure. They are usually characterized by spiraling winds that rotate **anti-clockwise** in the Northern Hemisphere and **clockwise** in the Southern hemisphere. Tropical cyclones are characterized by large pressure gradients. *The centre of the cyclone is mostly a warm, low-pressure, cloudless core known as the eye of the storm. Generally, the isobars are closely placed to each other, showing high-pressure gradients.

Under the influence of gravity, a landslide is an activity involving the sliding of rocks and soil along slopes.

*The frequency of **landslides** has been increasing in the Himalayas. Its root causes are increased frequency of earthquakes as well as **man-made activities** such as the construction of roads and dams. L**arge scale mining** has been conducted for the construction of roads, dams, and minerals in the Himalayas in recent years. That is why the frequency of landslides has increased in the Himalayas. ***The first Disaster Management Training Institute in country** has been established in **Latur, Maharashtra.**

1. Which of the following coasts of India was worst affected by the 2004 Tsunami?

(a) Malabar Coast (b) Konkan Coast
(c) Coromandel Coast (d) Northern Circars Coast

U.P.R.O./A.R.O (Mains) 2014

Ans. (c)

On 26th December, 2004, the Tsunami, which originated from the Indian Ocean, severely affected, the Coromandel Coast of India. The Coromandel Coast is southeastern coast of the Indian subcontinent. The coastline runs from False Divi Point in the north to Kanyakumari in the south.

2. Which of the following coastal areas of India was affected by 'Hudhud Cyclone'?

(a) Andhra Pradesh Coast (b) Kerala Coast
(c) Chennai Coast (d) Bengal Coast

M.P.P.C.S. (Pre) 2016

Ans. (a)

The Hudhud cyclone struck the Andhra Pradesh Coast at Vishakhapatnam in October 2014. Hudhud caused extremely heavy rainfall with strong gale winds, leading to structural damage over northern Andhra Pradesh. The name Hudhud was suggested by Oman.

3. **Tsunami Warning Centre in India is located in**
(a) Chennai (b) Vishakhapatnam
(c) Hyderabad (d) Port Blair
U.P.P.C.S. (Mains) 2012

Ans. (c)

The Indian Tsunami Early Warning Centre (ITEWC) is established at the Indian National Centre for Ocean Information Science (INCOIS) in Hyderabad. Its operation started in october, 2007.

4. **India Meteorological Department is established at:**
(a) New Delhi (b) Nagpur
(c) Jodhpur (d) Pune
U.P.P.C.S. (Mains) 2012

Ans. (a)

The India Meteorological Department was established in 1875 under the Ministry of Earth Sciences, with its Head Office situated at Lodhi Road in New Delhi. The Indian Institute of Tropical Meteorology and the deputy director's office of IMD (India Meteorological Department) are situated in Pune. Kolkata was the first head office of the India Meteorological Department. The headquarters of IMD was later shifted to Shimla in 1905, then to Pune in 1928, and to Delhi in 1944.

5. **Match List-I (Natural Hazards) with List-II (Regions) and select the correct answer using the codes given below the lists:**

List-I (Natural Hazards)	List-II (Regions)
A. Floods	**1. Himalayan foothill Zone**
B. Earthquakes	**2. Jharkhand and Northern Odisha**
C. Droughts	**3. Plains of Uttar Pradesh and Bihar**
D. Cyclones	**4. Mid-Eastern India.**

Code :

	A	B	C	D
(a)	3	1	2	4
(b)	3	1	4	2
(c)	2	3	1	4
(d)	4	2	3	1

47th B.P.S.C. (Pre) 2005

Ans. (b)

"Flood Forecast & Warning Organisation" was set up by Central Water Commission in 1969 to establish forecasting sites on Inter-State rivers at various flood-prone places in the country. The "National Flood Forecasting and Warning Network" of Central Water Commission, which comprised of 338 flood forecasting sites. The most drought-affected area is middle eastern India. Generally, the Cyclones affecteastern coast of India. Central eastern India is most affected by cyclones originating in the Bay of Bengal. Cyclones are generated more frequently in Odisha.

6. **Given below are two statements. One is labelled as Assertion and other as Reason :**
Assertion (A) : The frequency of landslides has increased in the Himalayas.
Reason (R) : There has been large scale mining in the Himalayas in recent years.
In the context of the above which one of the following is correct :
Code :
(a) Both (A) and (R) are true, and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true
U.P.P.C.S. (Mains) 2003

Ans. (a)

The frequency of landslides has increased in the Himalayas. Its root cause is the increased frequency of earthquakes as well as manual activities such as the construction of roads and dams. There has been large scale mining done for the construction of roads, dams, and minerals in the Himalayas in recent years. That is why the frequency of landslides has increased in the Himalayas. Thus, both (A) and (R) are true, and (R) is the correct explanation of (A).

7. **Cyclones are more frequent in the coastal areas of Bay of Bengal, because -**
(a) High temperatures in the Bay of Bengal
(b) Water in the Bay of Bengal has chemicals which help in the formation of cyclones
(c) Long chain of the island of Andaman and Nicobar acts as launching pad for cyclones
(d) The conical shape of Bay of Bengal funnels cyclones North wards when they are formed in the sea
M.P.P.C.S. (Pre) 1996

Ans. (a)

A cyclone is a large scale air mass that rotates around a strong centre of low pressure. They are usually characterized by inward spiraling winds that rotate counterclockwise in the northern hemisphere and clockwise in the southern hemisphere.
The cyclones affect the Bay of Bengal and Arabian Sea. More cyclones are formed in the Bay of Bengal due to low pressure, generated by high temperatures. These cyclones move in an anticlockwise direction. There are two definite seasons of tropical cyclones in the North Indian Ocean. One is from May to June, and the other is from mid-September to mid-December. May, June, October, and November are known for severe storms. The entire east coast from Odisha to Tamil Nadu is vulnerable to cyclones with varying frequency and intensity. Hence, Statement (a) is correct.

8. Assertion (A) : East coast is most prone to cyclones than western coast.
Reason (R) : Easter coast of India lies in the zone of north-east trade winds.
In the context of above statements, which of the following is correct?
Code :
(a) (A) and (R) both are true, and (R) is correct explanation of (A).
(b) (A) and (R) both are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (R) is true, but (A) is false.

U.P.P.C.S. (Pre) 2003

Ans. (b)

The eastern coast is more prone to cyclones than western coast, as cyclones originating in the Bay of Bengal move in an anticlockwise direction and affect whole east coast. Hence, the statement (A) is correct. The eastern coast of India lies in zone of northeast trade winds. Thus, both (A) and (R) are true and (R) is not the correct explanation of (A).

9. Assertion (A) : The Koyna region of Maharashtra is likely to become more earthquake-prone in near future.
Reason (R) : The Koyna dam is located on an old fault-plane which may get activated more frequently with changes in water-level in Koyna reservoir.
Select the correct answer using the code given below:
Code :
(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not a correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.P.C.S.(Pre) 2001

Ans. (a)

According to geologists, the Koyna dam is located on an old fault plane, which may get activated more frequently with changes in the water level in Koyna reservoir. Koyna region of Maharashtra may thus become more earthquake prone in the near future.

10. Which among the following states faces a maximum natural disaster?
(a) Andhra Pradesh (b) Odisha
(c) Bihar (d) Gujarat

U.P. Lower Sub. (Pre) 2002

Ans. (b)

Since Odisha is located on the coast of the Bay of Bengal, it faces floods and droughts more frequently due to cyclones and anti-cyclones. Thus, Odisha is affected by cyclones, floods and droughts. Therefore, it faces more natural disasters than other states.

11. The first Disaster Management Training Institution of the country is being established at:
(a) Hyderabad (Andhra Pradesh)
(b) Bangalore (Karnataka)
(c) Latur (Maharashtra)
(d) Chennai (Tamil Nadu)

U.P. Lower Sub. (Pre) 2004

Ans. (c)

The first Disaster Management Training Institute of the country has been established in Latur, Maharashtra by the CRPF. Notably, India's first disaster management training institute was set up on 29 April, 1957 in Nagpur. Since Nagpur is not given in the options, the question is based on the institute set up by the CRPF.

12. Which one of the following areas of India does not come under the zone of high seismic intensity ?
(a) Uttarakhand (b) Karnataka Plateau

(c) Kutch (d) Himachal Pradesh

Uttarakhand P.C.S. (Pre) 2010

U.P.P.C.S. (Pre) 2006

Ans. (b)

On the basis of data provided by India Meteorological Department and other agencies, Bureau of Indian Standards has published the latest version of seismic zoning map of India. The earthquake resistant design code of India assigns four levels of seismicity zone factors for India. India is divided into four seismic zones on the basis of earthquake intensity- namely zone 2, zone 3, zone 4 and zone 5.

The intensity of earthquakes in different zones:

Seismic Zone	Intensity on Modified Mercalli scale
Zone II (Low-intensity zone)	VI (or less) low damage
Zone III (Moderate intensity zone)	VII damage to buildings
Zone IV (Severe intensity zone)	VIII destruction of buildings
Zone V (Very severe intensity zone)	IX (and above) very high damage

Zone V: Zone V covers 10.9% of the area of the country with the highest risk suffering earthquakes of intensity MSK IX or greater. The IS code assigns a zone factor of 0.36 for Zone V. Structural designers use this factor for the earthquake resistant design of structures in Zone V. The zone factor of 0.36 is indicative of effective (zero period) level earthquake in this zone. It is referred to as the very high damage risk zone. The regions of J & K (UT), Ladakh (UT), Some parts of Uttarakhand and Bihar, the North-East Indian region and the Rann of Kutch fall in this zone.

Zone IV: This zone is called the High Damage Risk Zone and covers the area liable to MSK VIII. The IS code assigns a zone factor of 0.24 for Zone IV. The Indo-Gangetic basin and the capital of the country (Delhi) as well as the rest of the Himalayas is included in Zone IV. In Maharashtra, the Patan area (Koyananagar) is also in Zone IV. This zone covers the remaining parts of Himachal Pradesh, Sikkim, the northern parts of Uttar Pradesh, Bihar and West Bengal, parts of Gujarat and small portions of Maharashtra near the west coast and Rajasthan. It comprises 17.3% of the areas of the country.

Zone III: The IS code assigns a zone factor of 0.16 for Zone III. It covers the Lakshadweep islands. This zone is a classified as Moderate Damage Risk Zone. It includes 30.4% of the areas of the country.

Zone II: The IS code assigns a zone factor of 0.10 (maximum horizontal acceleration that can be experienced by a structure in this zone is 10% of gravitational acceleration) for zone II. It comprises 41.4 % of the areas of the country.

Major parts of the peninsular region and Karnataka Plateau fall in this Zone. Thus, it does not come under the zone of high seismic intensity.

13. India has been divided into how many Seismic Risk Zones?

(a) 5 (b) 6

(c) 4 (d) 7

U.P. Lower Sub. (Pre) 2013

Ans. (c)

See the explanation of the above question.

The commission held option (a) as the correct answer.

14. Which one of the following is not correctly matched ?

Cities		**Seismic Zones**
(a) Bhuj	–	IV
(b) Hyderabad	–	I
(c) Srinagar	–	V
(d) Chennai	–	II

U.P.P.S.C. (R.I.) 2014

Ans. (*)

Options (a), (b) & (d) are not correctly matched because Bhuj lies in seismic Zone V and Hyderabad falls in Seismic Zone II. Other options are correctly matched.

Cities		**Seismic Zones**
Bhuj	–	Zone V
Hyderabad	–	Zone II
Srinagar	–	Zone V
Chennai	–	Zone III

15. Assertion (A) : The frequency of floods in North Indian plains has increased during the last couple of decades.

Reason (R) : There has been a reduction in the depth of river valleys due to deposition of silt.

Code :

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A)

(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A)

(c) (A) is true, but (R) is false.

(d) (A) is false, but (R) is true.

I.A.S. (Pre) 2000

Ans. (a)

The frequency of floods in the North Indian plains has increased during the last couple of decades. Its main reason is the shallowness of rivers due to deposition of silt. There has been a reduction in the depth of river valleys due to deposition of silts, which causes flooding during normal rain.

16. The most flood-prone state of India is :

(a) Assam (b) Andhra Pradesh

(c) Bihar (d) Uttar Pradesh

U.P.P.C.S. (Pre) 2000

Ans. (c)

The most flood prone state of India is Bihar.

17. Among the following area of Uttar Pradesh which is maximum flood affected ?

(a) Western area

(b) Eastern area

(c) Middle area

(d) Northern area

U.P.P.C.S. (Mains) 2012

Ans. (b)

Uttar Pradesh is divided into high flood-affected, medium flood affected and low flood affected regions. High flood-affected region covers 48% of total flood-affected regions of Uttar Pradesh. Eastern areas are the most flood-affected regions of Uttar Pradesh.

18. Which of the following statements are true?

1. Natural disasters cause maximum damage in developing countries.

2. Bhopal gas tragedy was man-made.

3. India is a disaster free country.

4. Mangroves reduce the impact of cyclones.

Select the correct answer from the codes given below:

(a) 1, 2 and 3 (b) 2, 3 and 4

(c) 1, 2 and 4 (d) 1, 3 and 4

U.P. P.C.S. (Pre) 2018

Ans. (c)

Most of the damage due to natural disasters occurs in developing countries. The Bhopal gas tragedy was a result of human error. It is noteworthy that on December 2-3, 1984, thousands (about 5200) people died due to leakage of poisonous gas called Methyl Isocyanate (MIC) in the pesticide plant of Union Carbide India Limited, Bhopal.

Mangrove forests reduce the effect of cyclones due to their strong and spreading roots. India is a very sensitive/ vulnerable country to natural calamities. UNDP has also declared India as one of the 10 most disaster-prone countries in the world. It is mentioned in statement 3 that India is a disaster free country, which is false. Hence, it is clear that statements 1,2 and 4 are correct, while statement 3 is false.

19. Match List- I with List- II and select the correct answer using the code given below :

List- I (Natural Disaster)	**List- II (Affected Area/Region)**
A. Flood	**i. Himalayan Zone**
B. Earthquake	**ii. Plains of Uttar Pradesh and Bihar**
C. Drought	**iii. West and Central India Zone**
D. Tsunami	**iv. Southern Coastal area of India**

Code :

	A	B	C	D
(a)	ii	i	iii	iv
(b)	i	ii	iii	iv
(c)	iv	i	ii	iii
(d)	iii	i	ii	iv

M.P.P.C.S. (Pre) 2019

Ans. (a)

List- I **(Natural Disaster)**		List- II **(Affected Area/Region)**
Flood	-	Plains of Uttar Pradesh and Bihar
Earthquake	-	Himalayan Zone
Drought	-	West and Central India Zone
Tsunami	-	Southern Coastal area of India

20. Cyclone "Tauktae" was formed in which ocean?

(a) Bay of Bengal

(b) Indian Ocean

(c) Caspian Sea

(d) Arabian Sea

U.P.R.O./A.R.O. (Pre) 2021

Ans. (d)

Cyclone "Tauktae" originated in the Arabian Sea. This storm hit the coast of Gujarat in May 2021.